ERG Analysis API

Validation Report

Version 2.4.0

Report Date	18 June 2026
Pipeline Version	v2.4.0
Previous Version	v2.3.2 (validated 16 June 2026, 13/13 synthetic pass)
Organisation	AI Fairness CIO — Charity Commission No. 1218464
Prepared by	Hamid Tavakoli, Author & Project Manager
Repository	github.com/AI-Fairness-com/ERG-Analysis-API
OSF Pre-registration	https://doi.org/10.17605/OSF.IO/6WA42
Normative Reference	Baker et al. (2025). Doc Ophthalmol 150(2):47–64. DOI: 10.1007/s10633-025-10009-2

✓ VALIDATION RESULT: PASS

v2.4.0 validated against two demo datasets. All five new features confirmed functional.

1. Scope and Purpose

This document records the functional validation of ERG Analysis API v2.4.0, released 18 June 2026. v2.4.0 is an incremental release adding PhNR (photopic negative response) amplitude display for the LA 3 protocol, pipeline and HTML file renaming, and consistent version labelling throughout all outputs.

v2.4.0 does not introduce new normative reference data, new Z-score thresholds, new classification logic, or changes to the signal processing pipeline. All locked parameters from v2.3.2 are preserved unchanged.

1.1 Locked Parameters (Inherited from v2.3.2)

Parameter	Locked Value
Sampling rate (fs)	2000 Hz
Notch filter Q factor	50
Figure DPI	220
Stage 1 classifier	Binary: Normal / Abnormal
Stage 2 disease classes	DR, Glaucoma, AMD, Retinitis Pigmentosa / Rod Dystrophy, Cone Dystrophy, Complete CSNB
Total output classes	7 (1 Normal + 6 disease)
RP class name	Retinitis Pigmentosa / Rod Dystrophy (never standalone)
Normative reference	Baker et al. (2025) N=407
Stage 1 AUC target	≥0.94 (pre-registered aspirational)
Stage 2 macro-AUC target	≥0.80 (pre-registered aspirational)
OSF DOI	https://doi.org/10.17605/OSF.IO/6WA42
Citation style	Harvard author-date

2. Changes in v2.4.0

2.1 New Features

ID	Component	Description
Py-2	erg_v2_4_0.py	phnr_amp_uv added to layer_2_clinical_summary (LA 3 only; None for all other protocols)
Py-3	erg_v2_4_0.py	phnr_amp_uv added to layer_4_technical_audit (LA 3 only; None for all other protocols)
Py-4	erg_v2_4_0.py	PhNR raw amplitude component added to FHIR bundle for LA 3 protocol; note: "Raw amplitude only. Reference range under development. No Z-score classification applied (v2.4.0)."
HTML-3	ERG_API_v2_4.html	Sixth metric card added to result panel (id: resPhnrCard); hidden by default; shown only when protocol = LA 3
HTML-4	ERG_API_v2_4.html	JavaScript added to renderResult() to populate PhNR card and control visibility
HTML-5	ERG_API_v2_4.html	PhNR line added to Layer 2 section of .txt report download: PhNR amplitude : −X.X μV [LA 3 — ref. range under development]

2.2 File Renames and Version Updates

ID	File	Change
Py-1a	erg_v2_4_0.py	PIPELINE_VERSION: "2.3.2" → "2.4.0"; VERSION_DATE: "2026-05-15" → "2026-06-18"
Py-1b	erg_v2_4_0.py	Cell 6 print statement updated to v2.4.0
Py-1c	erg_v2_4_0.py	FHIR performer display string updated to v2.4.0
RENAME	erg_v2_4_0.py	Renamed from erg_v2_3_2.py; file header docstring updated to ERG_V2_4_0.ipynb
RENAME	ERG_API_v2_4.html	Renamed from ERG_API_updated.html
HTML-1	ERG_API_v2_4.html	Report text header updated: v2.3.2 → v2.4.0
HTML-2	ERG_API_v2_4.html	CSS grid: repeat(5, 1fr) → repeat(6, 1fr) to accommodate PhNR card
Srv	erg_server.py	PIPELINE_PATH updated to erg_v2_4_0.py; HTML_PATH updated to ERG_API_v2_4.html with fallback
Docs	README.md	All v2.3.2 version references updated to v2.4.0; v2.4.0 entry added to version history
Docs	CHANGELOG.md	v2.4.0 entry added; streaming filter future version corrected from v2.4.0 to v2.5.0

2.3 PhNR Sign Convention

The PhNR trough is stored internally in **phnr_amp_uv** as a positive absolute value per the ISCEV baseline-to-trough (BT) measurement convention (Frishman et al. 2018). Display in the result panel and .txt report prepends a minus sign (−) to maintain visual consistency with the a-wave negative sign convention. The FHIR bundle retains the raw positive absolute value with an explanatory note.

2.4 Deferred to v2.5.0

PhNR Z-score and traffic light classification were explicitly deferred. A systematic review of five PhNR literature sources (Frishman et al. 2018; Sarossy et al. 2021; Machida 2012; Prencipe et al. 2020; Colotto et al. 2000) found no published age-stratified normative means and standard deviations for full-field PhNR using gold foil electrode and broadband LA 3 (white) stimulus. The ISCEV extended protocol (Frishman et al. 2018) explicitly requires locally established normative data. Encoding approximate cross-study norms with different stimuli and electrodes was assessed as clinically unsafe and rejected.

3. Validation Runs

Two functional validation runs were performed on 18 June 2026 using the deployed local server at http://localhost:8080. Both demo datasets were previously generated and validated under v2.3.2.

3.1 Validation Run 1 — Early DR Demo (DA 3 Protocol)

Test parameters

CSV file	DEMO_DR_early_DA3_gold_foil_le35.csv
Protocol	DA 3
Electrode	Gold Foil
Age group	≤35y
Pre-stimulus	50 ms
Flash duration	1 ms
Notch filter	Off

Pipeline output

Parameter	Value	Z-score	Signal	Expected
A-wave amplitude	−134.7 μV	Z = −1.33	**GREEN**	GREEN ☑
B-wave amplitude	163.4 μV	Z = −2.22	**AMBER**	AMBER ☑
B-wave implicit time	56.5 ms	Z = +1.60	**GREEN**	Positive ☑
B/A ratio	1.21	Z = −3.39	**RED**	RED ☑
PhNR card	Not shown	—	—	Hidden (DA 3) ☑
Overall	—	—	**RED**	RED ☑

Run 1 result: PASS — PhNR card correctly hidden for DA 3 protocol; traffic light RED driven by B/A ratio Z = −3.39

3.2 Validation Run 2 — Early Glaucoma Demo (LA 3 Protocol)

Test parameters

CSV file	DEMO_Glaucoma_early_LA3_gold_foil_36to59.csv
Protocol	LA 3
Electrode	Gold Foil
Age group	36–59y

Pre-stimulus	50 ms
Flash duration	1 ms
Notch filter	Off

Pipeline output

Parameter	Value	Z-score	Signal	Expected
A-wave amplitude	−4.1 μV	Z = −3.10	**RED**	RED ☑
B-wave amplitude	102.8 μV	Z = −0.22	**GREEN**	GREEN ☑
B-wave implicit time	33.5 ms	Z = +4.05	**RED**	RED ☑
B/A ratio	25.04	Z = N/A	—	N/A (LA 3) ☑
PhNR amplitude	−9.3 μV	None	**Pending**	Shown, no Z ☑
PhNR card visible	Yes	—	—	LA 3 shown ☑
PhNR FHIR component	9.3 uV + note	—	—	Present ☑
PhNR .txt report	−9.3 μV	—	—	Correct sign ☑
Overall	—	—	**RED**	RED ☑

Run 2 result: PASS — PhNR card shown with −9.3 μV; no Z-score; traffic light RED driven by b-wave implicit time Z = +4.05 only

4. v2.4.0 Release Checklist

#	Item	Status
1	PhNR card appears on LA 3, hidden on all other protocols	✅ PASS
2	PhNR value displayed as negative μV (−9.3 μV)	✅ PASS
3	PhNR sub-label reads “Ref. range pending”	✅ PASS
4	No Z-score assigned to PhNR	✅ PASS
5	No traffic light colour applied to PhNR	✅ PASS
6	Overall traffic light driven exclusively by validated Z-scores	✅ PASS
7	PhNR component present in FHIR bundle with correct note	✅ PASS
8	PhNR line in Layer 2 of .txt report with pending note	✅ PASS
9	Pipeline version consistently v2.4.0 in all outputs (FHIR, .txt, HTML, badge)	✅ PASS
10	fs = 2000 Hz locked throughout	✅ PASS
11	Python syntax check passed (ast.parse)	✅ PASS
12	Pipeline file named erg_v2_4_0.py	✅ PASS
13	HTML file named ERG_API_v2_4.html	✅ PASS
14	erg_server.py references updated filenames with fallback	✅ PASS
15	README.md and CHANGELOG.md updated and pushed to repository	✅ PASS

5. Inherited Validation from v2.3.2

v2.4.0 does not modify any signal processing, feature extraction, Z-score computation, or normative reference data. All 13/13 synthetic validation scenarios from v2.3.2 are inherited without re-execution. The v2.3.2 validation report (**docs/VALIDATION_REPORT_v2.3.2.docx**) remains the authoritative synthetic validation document.

Validation type	Dataset	Result	Status
Synthetic (v2.3.2, inherited)	12 scenarios × 5 protocols × 4 electrode types	**13/13**	✅ PASS
Technical (external)	OculusGraphy 2020 (n=149)	**100%**	✅ PASS
Normative integration	Baker et al. 2025 (N=407) — all 48 μ/σ values verified <0.02	**48/48**	✅ PASS
Functional v2.4.0 — Run 1	Early DR demo (DA 3, Gold Foil, ≤35y)	PASS	✅ PASS
Functional v2.4.0 — Run 2	Early Glaucoma demo (LA 3, Gold Foil, 36–59y)	PASS	✅ PASS
Sensitivity (real pathology)	Planned for V3.0	—	⏳ PENDING

6. Known Limitations and Deferred Items

ID	Description	Severity	Target
L1	PhNR Z-score and traffic light classification not implemented — no validated normative dataset available for Gold Foil + broadband LA 3 (Frishman et al. 2018)	Medium	v2.5.0
L2	B/A ratio displayed as raw number for LA 3 (physiologically meaningless) — cosmetic pre-existing issue from v2.3.2	Low	v2.5.0
L3	DTL reference ranges identical to Gold Foil — Deming regression transformation from Baker 2025 Table 2 not yet applied	Medium	v2.5.0
L4	Signal orientation auto-detection not implemented — inverted recordings not corrected automatically	Low	v2.5.0
L5	Baker 2025 normative data hardcoded — not externalised to JSON/YAML with versioning	Low	v2.5.0
L6	No unit test framework — pytest suite not yet created	Medium	v2.5.0
L7	SHAP Level 2 (spectrogram) pending — requires trained ViT model	Low	v3.0

7. References

Baker RA, Leo SM, Clowes WIN, et al. ISCEV standard full-field ERG reference limits from 407 healthy subjects, derived from transference and validation of reference data between electrode types and centres. Documenta Ophthalmologica. 2025;150(2):47–64. DOI: 10.1007/s10633-025-10009-2

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